

Probability - Homework 1

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1.2-17

(a)

To draw 4 cards with the same face value, there's 13 possibilities, then draw one from the rest of cards, there's $52 - 4 = 48$ possibilities (By definition, it's impossible to have same value with the previous 4 cards.), Hence, the probability is:

$$\frac{13 \times 48}{\binom{52}{4}} = \frac{1}{4165}$$

(b)

We can choose two type of from 13 face values, and for each face value we have 4 colors to choose, then we can choose either 2 or 3 cards from them (but not for both types). (btw, we should time a extra 2, for instance, if we need to construct a full-house using 2 and 3, we can draw 2 cards from face value 2 and 3 cards from face value 3, or draw 3 cards from face value 2 and 2 cards from face value 3, there's two possibilities.) Thus, the probability is:

$$\frac{\binom{13}{2} \binom{4}{2} \binom{4}{3} \times 2}{\binom{52}{5}} = \frac{6}{4165}$$

(c)

Choose 3 ranks from 13 ($\binom{13}{3}$), then decide which of the 3 ranks forms the triple ($\times 3$, since any of the 3 chosen ranks can be assigned the triple role), choose 3 suits for it ($\binom{4}{3}$), and choose 1 suit for each of the remaining 2 kicker ranks ($\binom{4}{1}^2$). Thus, the probability is:

$$\frac{\binom{13}{3} \binom{4}{3} (\binom{4}{1})^2 \times 3}{\binom{52}{5}} = \frac{88}{4165}$$

(d)

Choose 3 ranks from 13 ($\binom{13}{3}$), then decide which of the 3 ranks is the single kicker ($\times 3$, since any of the 3 chosen ranks can be assigned the kicker role), choose 1 suit for it ($\binom{4}{1}$), and choose 2 suits for each of the two pairs ($\binom{4}{2}^2$). Thus, the probability is:

$$\frac{\binom{13}{3} \binom{4}{1} (\binom{4}{2})^2 \times 3}{\binom{52}{5}} = \frac{198}{4165}$$

(e)

Choose 4 ranks from 13 ($\binom{13}{4}$), then decide which of the 4 ranks forms the pair ($\times 4$, since any of the 4 chosen ranks can be assigned the pair role), choose 2 suits for it ($\binom{4}{2}$), and choose 1 suit for each of the 3 remaining kicker ranks ($\binom{4}{1}^3$). Thus, the probability is:

$$\frac{\binom{13}{4} \binom{4}{2} (\binom{4}{1})^3 \times 4}{\binom{52}{5}} = \frac{352}{833}$$

1.3-13**(a)****(b)****(c)****(d)****(e)****1.4-15**

Define the random variable X as: The fourth white balls appeaing at X -th round. And P be the probability function.

(a)

With replacement, the probability of taking white balls p is $\frac{1}{2}$. Also we can consider the experiment as a bernoulli trial.

$$P(X = 4) = (1 - p)^3 p = \frac{1}{16}$$

$$P(X = 5) = (1 - p)^4 p = \frac{1}{32}$$

$$P(X = 6) = (1 - p)^5 p = \frac{1}{64}$$

$$P(X = 7) = (1 - p)^6 p = \frac{1}{128}$$

(b)

We can consider the results as the permutations of 10 white balls and 10 red balls. And if we take d -th white ball at r -th round (where $d \leq r$), the number of permutations is:

$$\frac{(r-1)!}{(d-1)!(r-d)!} \times \frac{(20-r)!}{(10-d)!(10-r+d)!} = \binom{r-1}{d-1} \times \binom{20-r}{10-d}$$

The total number of permutations of the 20 balls is :

$$\frac{20!}{10!10!} = \binom{20}{10}$$

(c)

1.5-4

Consider 4 partitions (age 16-25, 26-50, 51-65, 66-90, respectively) as B_1 , B_2 , B_3 , B_4 , and A be the event of accident. By Bayes' Theorem we obtain the answer:

$$\begin{aligned} P(B_1|A) &= \frac{P(A|B_1)P(B_1)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) + P(A|B_4)P(B_4)} \\ &= \frac{0.05 \times 0.10}{0.05 \times 0.10 + 0.02 \times 0.55 + 0.03 \times 0.20 + 0.04 \times 0.15} \\ &= 0.178571428571 \approx 0.1786 \end{aligned}$$